Figure 1:

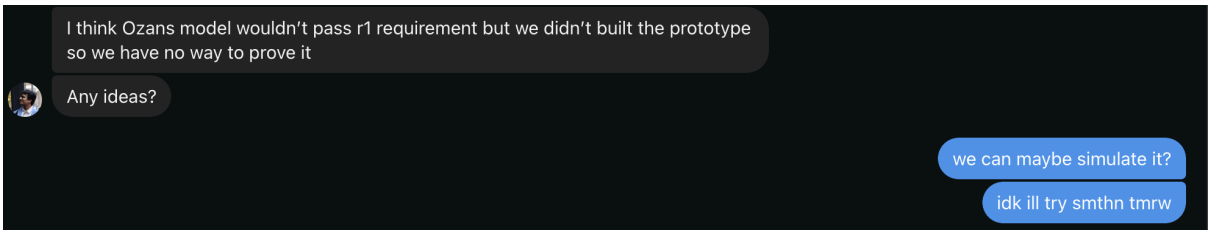
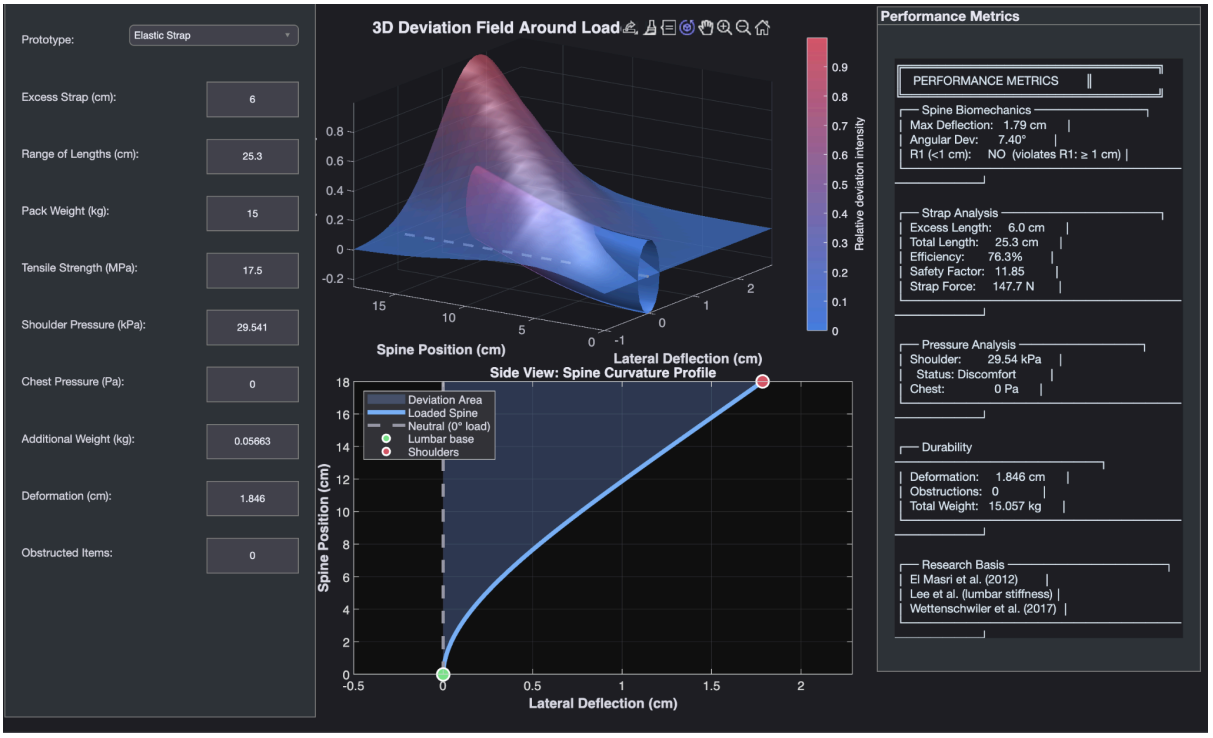


Figure 2:

Figure 3:


Prototype:
Elastic Strap
Excess Strap (cm):
6
Range of Lengths (cm):
25.3
Pack Weight (kg):
15
Tensile Strength (MPa):
17.5
Shoulder Pressure (kPa):
29.541
Chest Pressure (Pa):
0
Additional Weight (kg):
0.05663
Deformation (cm):
1.846
Obstructed Items:
0
3D Deviation Field Around Load
Spine Position (cm)
Lateral Deflection (cm)
Relative deviation intensity
Side View: Spine Curvature Profile
Deviation Area
Loaded Spine
Neutral (0° load)
Lumbar base
Shoulders
Spine Position (cm)
Lateral Deflection (cm)
Performance Metrics
PERFORMANCE METRICS
Spine Biomechanics
Max Deflection: 1.79 cm
Angular Dev: 7.40°
R1 (<1 cm): NO (violates R1: ≥ 1 cm)
Strap Analysis
Excess Length: 6.0 cm
Total Length: 25.3 cm
Efficiency: 76.3%
Safety Factor: 11.85
Strap Force: 147.7 N
Pressure Analysis
Shoulder: 29.54 kPa
Status: Discomfort
Chest: 0 Pa
Durability
Deformation: 1.846 cm
Obstructions: 0
Total Weight: 15.057 kg
Research Basis
El Masri et al. (2012)
Lee et al. (lumbar stiffness)
Wettenschwiler et al. (2017)